CS1231S Discrete Structures

WEEK 13 RECITATION
TUTORIAL 11 & REVISION

12 November 2026

Aaron Tan



CS1231S (AY2026/27 Sem1)
QnA



Selected Past Years' Exam Questions

- AY2022/23 Semester 2
  - Q15, Q16, Q20, Q23(f)
- AY2022/23 Semester 1
  - Q24, Q25
- AY2021/22 Semester 1
  - Q22
- AY2024/25 Semester 1
  - Q8

Past-Year's Exam Question!

AY2022/23 Semester 2

Hi prof, can you please explain Q15?

15. Which of the following are lattices?
    (i) The poset $(\mathbb{Z}^+ \setminus \mathbb{Z}_{>120}, |)$.
    (ii) The poset $(\mathcal{P}(\{a, b, c, d\}), \subseteq)$.
    (iii) The poset $(K, \preccurlyeq)$ in question 11.

A. Only (i).
B. Only (ii).
C. Only (i) and (ii).
D. Only (i) and (iii).
E. None of the options (A), (B), (C), (D) are correct.

Let $S \subseteq A$ in the poset $(A, \preccurlyeq)$.

Upper Bound:
If there exists an element $u \in A$ such that $s \preccurlyeq u$ for all $s \in S$, then u is called an *upper bound* of S.

Lower Bound:
If there exists an element $l \in A$ such that $l \preccurlyeq s$ for all $s \in S$, then l is called a *lower bound* of S.

Least Upper Bound:
If a is an upper bound of S such that $a \preccurlyeq u$ for all upper bound u of S, then a is the *least upper bound* of S, denoted by $lub(S)$.

Greatest Lower Bound:
If a is a lower bound of S such that $l \preccurlyeq a$ for all lower bound l of S, then a is the *greatest lower bound* of S, denoted by $glb(S)$.

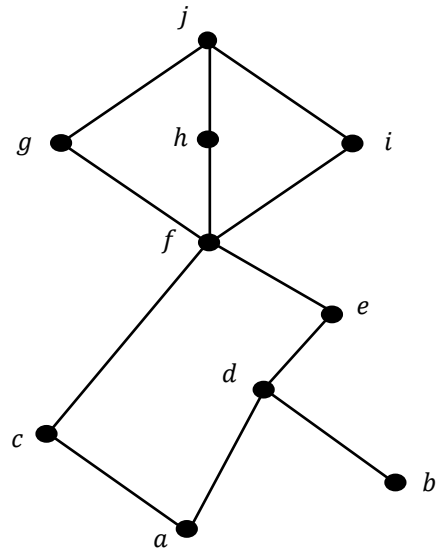
Lattice:
A poset in which every pair of elements has both a least upper bound and a greatest lower bound is called a *lattice*.

Past-Year's Exam Question!

AY2022/23 Semester 2

Hi prof, can you please explain Q15?

11. The Hasse diagram of a partial order $\preccurlyeq$ on a set $K = \{a, b, c, d, e, f, g, h, i, j\}$ is shown below.



15. Which of the following are lattices?
    (i) The poset $(\mathbb{Z}^+ \setminus \mathbb{Z}_{>120}, |)$.
    (ii) The poset $(\mathcal{P}(\{a, b, c, d\}), \subseteq)$.
    (iii) The poset $(K, \preccurlyeq)$ in question 11.

A. Only (i).
B. Only (ii).
C. Only (i) and (ii).
D. Only (i) and (iii).
E. None of the options (A), (B), (C), (D) are correct.

Past-Year's Exam Question!

AY2022/23 Semester 2

16. Given any non-empty finite set A, let $\mathcal{P}(A)$ be the power set of A and $\mathcal{C}$ any partition of A. Which of the following statements are always true?

    (i) $\exists S \in \mathcal{C}$ such that $S \subseteq \mathcal{P}(A)$.
    (ii) $\exists S \in \mathcal{P}(A)$ such that $S \in \mathcal{C}$
    (iii) $\exists S \in A$ such that $\{S\} \in \mathcal{C}$.

A. Only (i).
B. Only (ii).
C. Only (i) and (ii).
D. Only (i) and (iii).
E. None of the options (A), (B), (C), (D) are correct.

Past-Year's Exam Question!

AY2022/23 Semester 2

20. A **height-balanced binary tree** is a binary tree in which the height of the left subtree and the height of the right subtree of any vertex differ by at most one. (By convention, if a vertex does not have a left/right child, then the height of the absent left/right subtree is -1.)

    What is the minimum number of vertices in a height-balanced binary tree of height 4?

    A. 7
    B. 9
    C. 12
    D. 23
    E. None of the above.

Past-Year's Exam Question!

AY2022/23 Semester 2

23 (f)

Let $A = \{a, b, c, \cdots, z\}$ be the set of lower-case letters and S be the set of all strings of length 6 over A. Note that the letters in a string need not be distinct.

(i) Let $S_{1v} \subseteq S$ where each element in S_{1v} contains exactly one vowel. What is $|S_{1v}|$?
(ii) Let $S_{2v} \subseteq S$ where each element in S_{2v} contains exactly two vowels. What is $|S_{2v}|$?
(iii) Let $S_{\geq 1v} \subseteq S$ where each element in S_{1v} contains at least one vowel. What is $|S_{\geq 1v}|$?
(iv) Let $S_{\geq 2v} \subseteq S$ where each element in S_{2v} contains at least two vowels. What is $|S_{\geq 2v}|$?

(i)

Past-Year's Exam Question!

AY2022/23 Semester 2

23 (f)

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(ii)

Past-Year's Exam Question!

AY2022/23 Semester 2

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(iii) Let $S_{\geq 1v} \subseteq S$ where each element in S_{1v} contains at least one vowel. What is $|S_{\geq 1v}|$?
(iv) Let $S_{\geq 2v} \subseteq S$ where each element in S_{2v} contains at least two vowels. What is $|S_{\geq 2v}|$?

(iii)

(iv)

Past-Year's Exam Question!

AY2022/23 Semester 1

24(a) Let $f: \mathbb{R} \to \mathbb{R}$ and $g: \mathbb{R} \to \mathbb{R}$ be two bijections.
The function $h: \mathbb{R} \to \mathbb{R}$ is defined as follows:

$$h(x) = f(x) + g(x), \text{ for any } x \in \mathbb{R}.$$

Provide a counterexample to disprove this statement: h is a bijection.

Past-Year's Exam Question!

AY2022/23 Semester 1

24(b) Definitions:

A function $f: \mathbb{R} \to \mathbb{R}$ is said to be **increasing** iff $x_2 > x_1 \to f(x_2) > f(x_1), \forall x_1, x_2 \in \mathbb{R}$.

A function $f: \mathbb{R} \to \mathbb{R}$ is said to be **decreasing** iff $x_2 > x_1 \to f(x_2) < f(x_1), \forall x_1, x_2 \in \mathbb{R}$.

Is the following statement true or false? If true, prove it; if false, disprove it.

If g is an increasing function and h is a decreasing function,
then $h \circ g$ is a decreasing function.

Past-Year's Exam Question!

AY2022/23 Semester 1

24(c) A **totally ordered set** is a pair (A, R) where A is a set and R is a total order on A.
Define a relation $\sim$ on two totally ordered sets as follows:

> $(A, R_A) \sim (B, R_B)$ if and only if there exists a bijection $f: A \to B$ such that
> $\forall a_1, a_2 \in A, \quad a_1 R_A a_2 \leftrightarrow f(a_1) R_B f(a_2)$.

Prove that $(\mathbb{R}, \leq) \sim (\mathbb{R}, \geq)$, where $\leq$ and $\geq$ are the usual numerical less-than-or-equal-to and greater-than-or-equal-to symbols respectively.

Past-Year's Exam Question!

AY2022/23 Semester 1

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Past-Year's Exam Question!

AY2022/23 Semester 1

25. Let A be a set such that $|A \times A| = |A|$. Suppose there is a countably infinite subset B of A such that there is no surjection from B to A.

(a) Prove or disprove: "A is countable".

Past-Year's Exam Question!

AY2022/23 Semester 1

25. Let A be a set such that $|A \times A| = |A|$. Suppose there is a countably infinite subset B of A such that there is no surjection from B to A.

(b) Suppose S is a countable subset of $A \times A$. Prove that there exists some element $x \in A$ such that $\{x\} \times A \subseteq (A \times A) \setminus S$.

Past-Year's Exam Question!

AY2021/22 Semester 1

22(a) Must $|A_1 \cup A_2| \neq |B \cup Z|$ if A_1, A_2, B are uncountable sets and Z is a countable set?

Past-Year's Exam Question!

AY2021/22 Semester 1

22(b) Must $|C_1 \times C_2| = |D_1 \times D_2|$ if C_1, C_2, D_1, D_2 are all countable sets?

Past-Year's Exam Question!

AY2024/25 Semester 1

8. Which of the following statements are true about relations on a non-empty set?

   (i) The transitive closure of a reflexive relation is always symmetric.
   (ii) The reflexive closure of a transitive relation is always symmetric.
   (iii) It is not possible to have a symmetric closure of an anti−symmetric relation.

A. Only (iii).
B. Only (i) and (ii).
C. Only (ii) and (iii).
D. None of (i), (ii) and (iii).
E. None of options (A), (B), (C), (D) are correct.